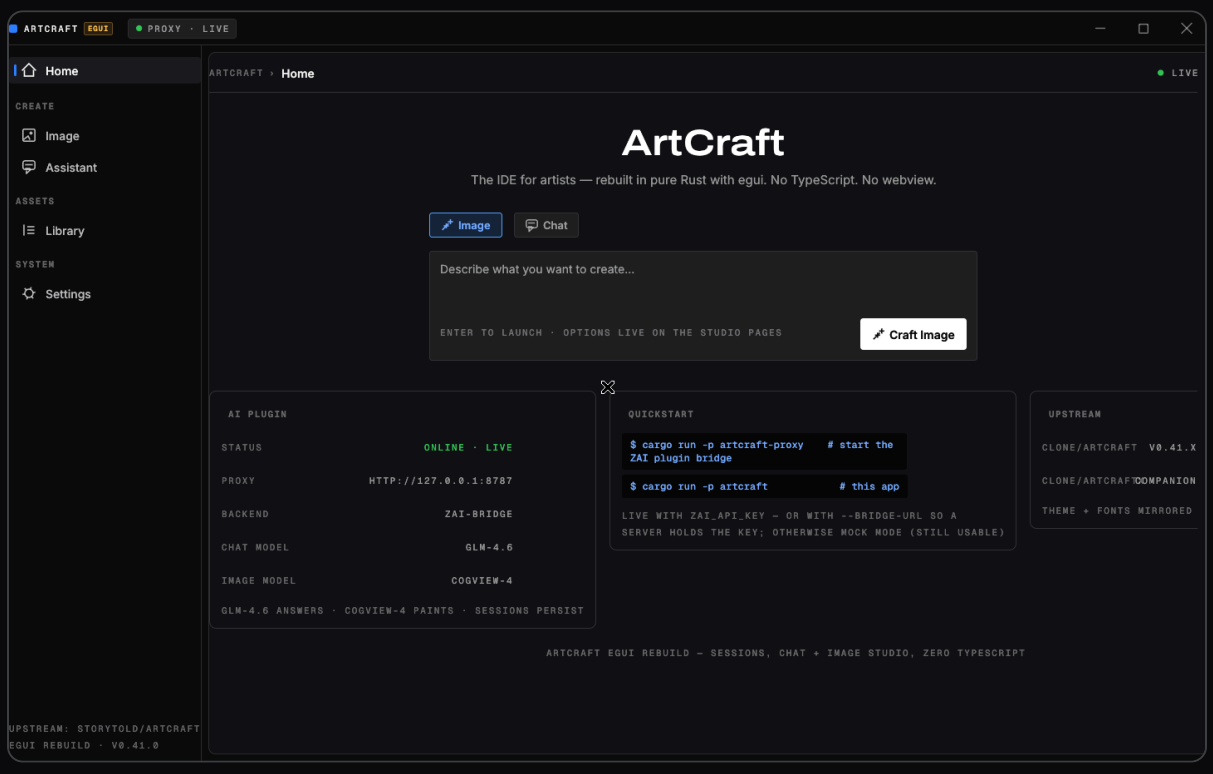
PROXY · LIVE — GLM-4.6 + COGVIEW-4

ArtCraft
design process

How the pure-egui rebuild of ArtCraft was tested end to end with real ZAI models — every prompt typed, every image generated, every screen captured from the running Linux build.

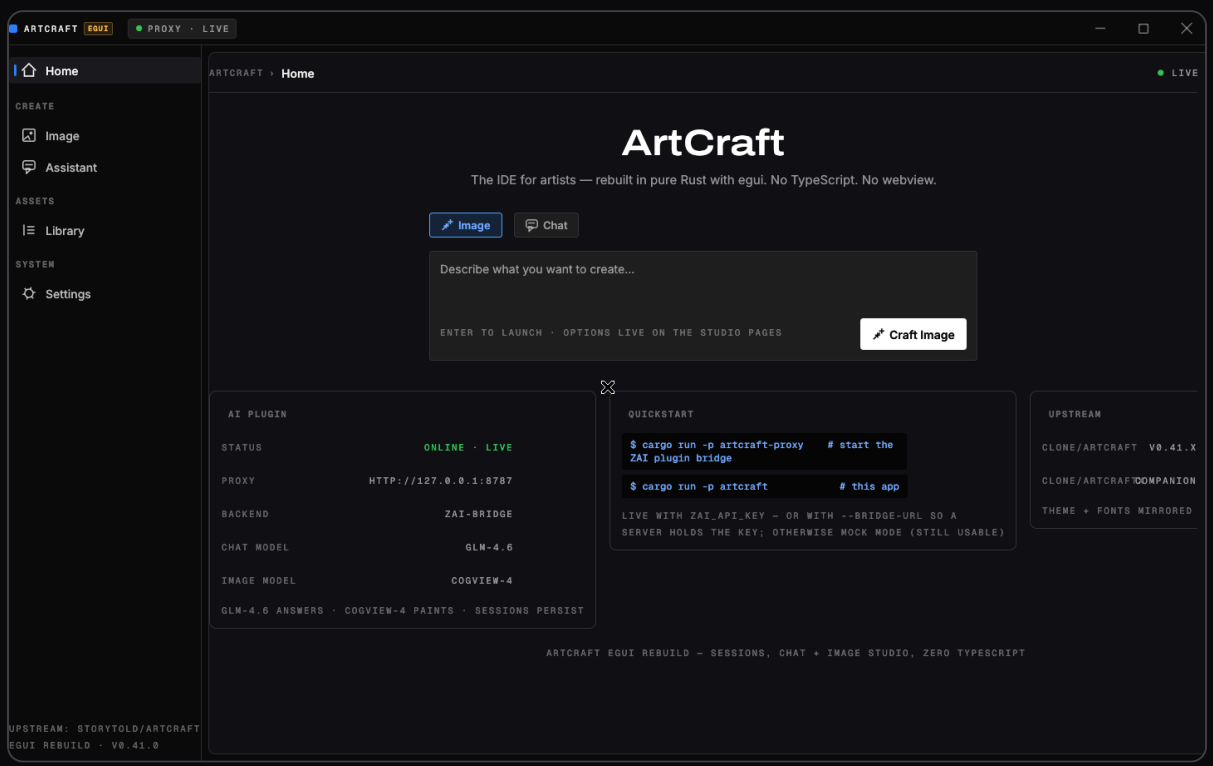
RUST + EGUI 0.36 | NO TYPESCRIPT | GLM-4.6 CHAT | COGVIEW-4 IMAGES | ZERO API KEYS



CHAPTER 01

The app at a glance

The rebuild ships as a native desktop app with the same brutalist dark shell as the TypeScript original: a 40 px title bar with a live proxy chip, a 232 px sidebar, a breadcrumb header and a rounded content card. The landing page mirrors the web app's hero — a big centered composer with **Image / Chat** mode chips, so the very first thing you can do is type. No API key, no login, no configuration: the app ships pointed at the local **artcraft-proxy** bridge, and the header dot turns green the moment real models answer.



HOME — HERO COMPOSER WITH IMAGE / CHAT CHIPS

STATUS CARDS: ONLINE · LIVE · ZAI-BRIDGE

Using the app is a three-move loop, and this document walks through all three with real captures:

01 **Brainstorm with the Assistant** — describe what you want; GLM-4.6 returns a polished, ready-to-paste image prompt.

02 **Craft it in the Image studio** — paste or type a prompt into the always-visible bottom composer; CogView-4 paints it and the result lands in the feed.

03 **Review and keep** — open the lightbox for a full-size look; every image is auto-saved to the Library on disk.

CHAPTER 02

Step 1 — brainstorm with GLM-4.6

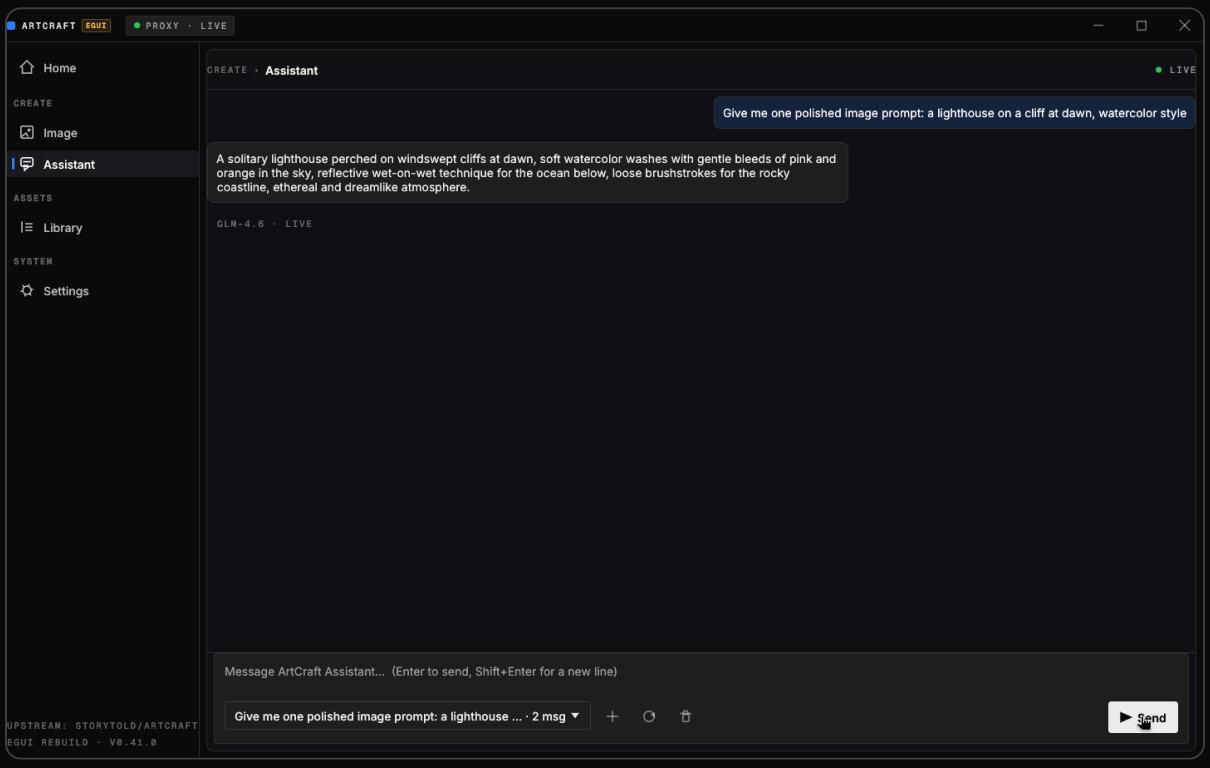
The Assistant page keeps the composer docked at the bottom — Enter sends, Shift+Enter breaks a line, and sessions are stored by the proxy with full history every turn. The first ask was deliberately open: **“Give me one polished image prompt: a lighthouse on a cliff at dawn, watercolor style.”** GLM answered with a single, production-ready prompt — no meta-commentary, nothing to strip out.

USER · SENT FROM THE DOCKED COMPOSER

Give me one polished image prompt: a lighthouse on a cliff at dawn, watercolor style

ARTCRAFT ASSISTANT · GLM-4.6 · LIVE

“A solitary lighthouse perched on rugged seaside cliffs at dawn, soft pastel watercolor washes blending with gentle brushstrokes, mist rising from the ocean below, warm golden light breaking through clouds, serene and ethereal atmosphere.”



ASSISTANT — REAL GLM-4.6 REPLY, BUBBLE HUGS THE TEXT SESSION PICKER: 2 MSG · LIVE

A second question doubled as a smoke test: **“Explain in two sentences how the image studio works in this app.”** The reply wrapped cleanly inside its bubble and

“what can this studio do?” mid-session. Long replies never run off the window; bubbles cap at 72 % of the row width and hug their text.

CHAPTER 03

Step 2 — CogView-4 paints it

GLM’s prompt went straight into the Image studio composer — typed exactly as the assistant wrote it, no edits. The Generate button (or just Enter) fired the request through the Rust proxy to CogView-4; about a minute later the painting appeared at the top of the feed with its model and size caption, and a toast confirmed it was saved to the Library.

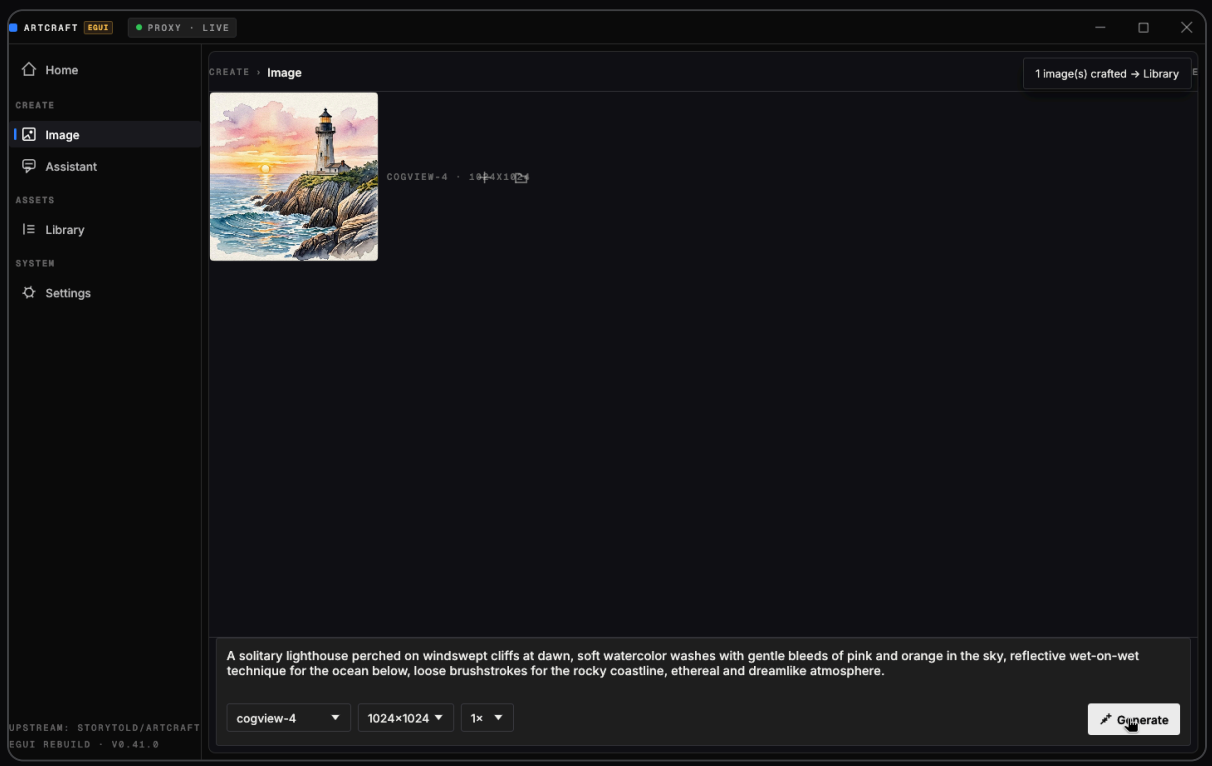


IMAGE STUDIO — GLM’S PROMPT IN THE DOCKED COMPOSER

“1 IMAGE(S) CRAFTED → LIBRARY” TOAST

PROMPT P1 · TYPED ON THE HOME HERO · BECAME THE STUDIO PAINTING

```
a painters studio at golden hour, oil on canvas style, warm light through tall windows, brushes and easels, soft shadows, high quality, detailed
```

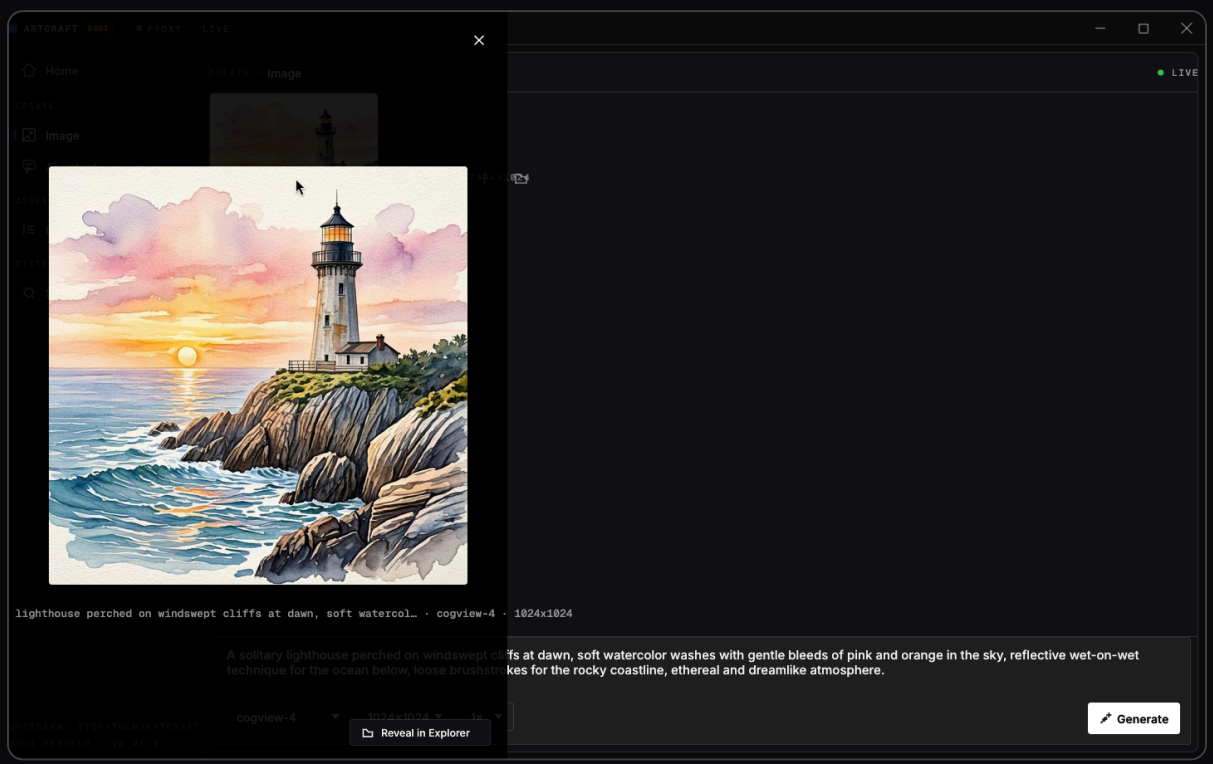
PROMPT P2 · WRITTEN BY GLM-4.6 · PAINTED BY COGVIEW-4

```
A solitary lighthouse perched on windswept cliffs at dawn, soft watercolor washes with gentle bleeds of pink and orange in the sky, reflective wet-on-wet technique for the ocean below, loose brushstrokes for the rocky coastline, ethereal and dreamlike atmosphere.
```

CHAPTER 04

Step 3 — lightbox & library

Clicking any tile opens the lightbox: the artwork at full size on a dimmed backdrop, with its caption (prompt prefix · model · size) and a **Reveal in Explorer** shortcut to the file on disk. Escape closes it. The Library page lists every generated image with its prompt, model and size, and persists across restarts via `artcraft-data/library.json`.



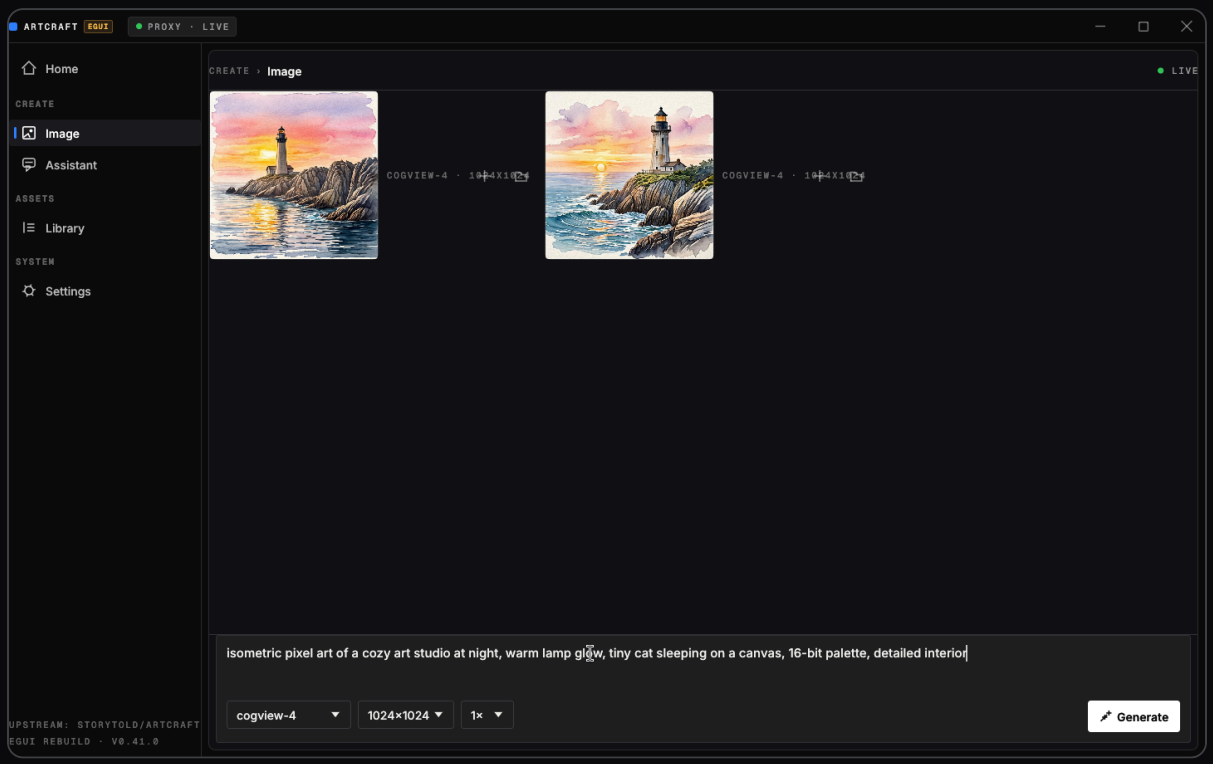
LIGHTBOX — FULL-SIZE REVIEW

COGVIEW-4 · 1024X1024

CHAPTER 05

Second pass — pixel art iteration

The design loop was run a second time with a different style to prove iteration: select-all, retype, generate. The pixel-art prompt below was typed into the composer (visible mid-edit in the capture) and CogView-4 rendered the cozy isometric studio — warm lamp glow, the sleeping cat on the canvas, 16-bit palette — in about 55 seconds.



ITERATION — NEW PROMPT TYPED INTO THE SAME COMPOSER

MODEL · SIZE · COUNT PICKERS BELOW

PROMPT P3 · TYPED FOR THE SECOND PASS

```
isometric pixel art of a cozy art studio at night, warm lamp glow, tiny cat sleeping on a canvas, 16-bit palette, detailed interior
```

HONEST ITERATION NOTE

One earlier attempt typed a fresh prompt without clearing the composer, and the caret landed mid-text — the request concatenated the lighthouse and pixel-art prompts. The model happily blended them into a calmer lighthouse at sunset (kept in the gallery as a happy accident). The app itself behaved correctly; the lesson is compositional, not architectural: clear the box, or reuse a previous prompt deliberately via the tile's **Reuse prompt** button.

CHAPTER 06

The gallery

All four artworks produced during this session, exactly as saved by the app to `artcraft-data/generated/`. Every image was generated live by CogView-4 through the proxy bridge during the GUI test run — none are mockups or placeholders.



01 · PAINTER'S STUDIO, GOLDEN HOUR — PROMPT P1



02 · LIGHTHOUSE WATERCOLOR — PROMPT P2



03 · BLENDED-PROMPT LIGHTHOUSE — THE HONEST ACCIDENT



04 · PIXEL-ART STUDIO WITH CAT — PROMPT P3

CHAPTER 07

Zero-key architecture, proven live

Neither the desktop app nor the Rust proxy ever holds a credential. Two live backends were verified during this run: the local **artcraft-proxy** on `127.0.0.1:8787` (which relays model calls to a server-side ZAI bridge), and the hosted backend reached by simply pasting a URL into Settings — the exact flow a Windows user repeats with the preview URL. The settings readout shows the truth in green: **mode live**, and the API key row reads **“none — held server-side by the ZAI bridge”**.

ARTCRAFT (EGUI)

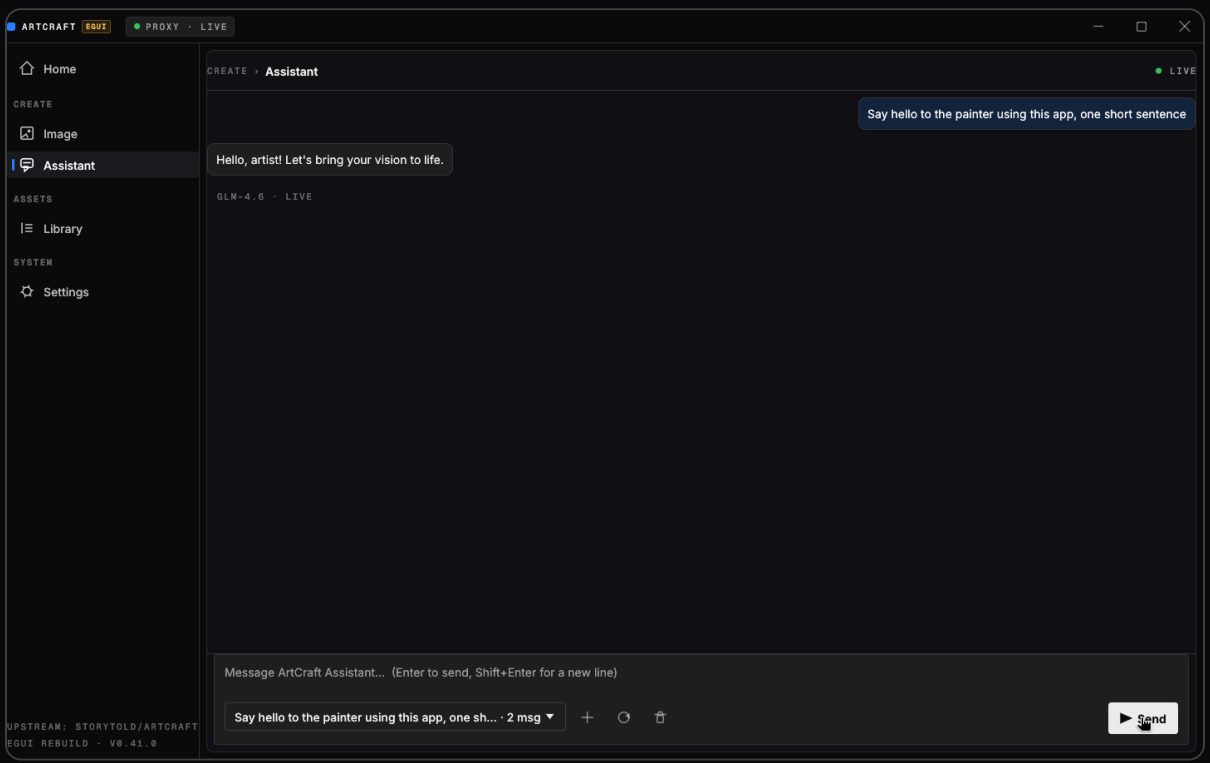
Settings → Proxy URL. Only a URL ever reaches the app.

ARTCRAFT-PROXY

Rust · sessions, chat & image endpoints on 127.0.0.1:8787.

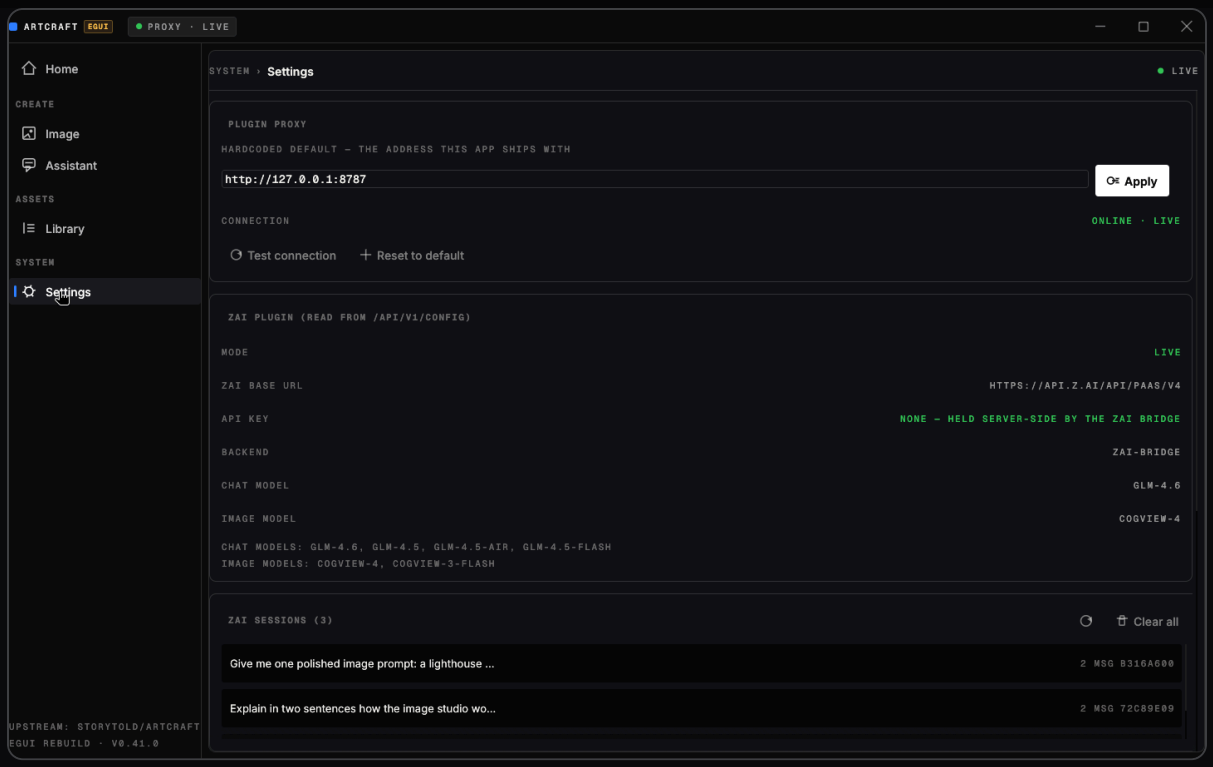
ZAI BRIDGE

Holds the credential; GLM-4.6 + CogView-4 answer here.



PASTE-URL FLOW — FRESH SESSION ON THE HOSTED BACKEND

“HELLO, ARTIST! LET’S BRING YOUR VISION TO LIFE.”



SETTINGS — LIVE MODE, NO KEY ON THIS MACHINE

BACKEND: ZAI-BRIDGE · GLM-4.6 · COGVIEW-4

The app also self-heals across backends: switching the Proxy URL drops the stale session id, and if a session ever vanishes the next message recreates it transparently. That single detail is what makes “paste a URL and go” trustworthy rather than fragile.

CHAPTER 08

Every prompt in this log

The complete, unabridged set of prompts and messages that produced this document — typed by hand or returned by GLM-4.6, all executed against live models.

P1 · HUMAN · HOME HERO COMPOSER · → STUDIO PAINTING

```
a painters studio at golden hour, oil on canvas style, warm light through tall windows, brushes and easels, soft shadows, high quality, detailed
```

C1 · HUMAN · ASSISTANT

```
Give me one polished image prompt: a lighthouse on a cliff at dawn, watercolor style
```

P2 · GLM-4.6 REPLY TO C1 · → WATERCOLOR LIGHTHOUSE

A solitary lighthouse perched on rugged seaside cliffs at dawn, soft pastel watercolor washes blending with gentle brushstrokes, mist rising from the ocean below, warm golden light breaking through clouds, serene and ethereal atmosphere.

P2′ · GLM-4.6 · SECOND SESSION VARIANT OF THE LIGHTHOUSE PROMPT

A solitary lighthouse perched on windswept cliffs at dawn, soft watercolor washes with gentle bleeds of pink and orange in the sky, reflective wet-on-wet technique for the ocean below, loose brushstrokes for the rocky coastline, ethereal and dreamlike atmosphere.

C2 · HUMAN · ASSISTANT SMOKE TEST

Explain in two sentences how the image studio works in this app

C3 · HUMAN · ASSISTANT (HOSTED BACKEND)

Say hello to the painter using this app, one short sentence

P3 · HUMAN · IMAGE STUDIO · → PIXEL-ART STUDIO

isometric pixel art of a cozy art studio at night, warm lamp glow, tiny cat sleeping on a canvas, 16-bit palette, detailed interior

Type a sentence.
Get a painting.

ArtCraft rebuilt in pure Rust with egui — sessions, chat and image studio, wired to ZAI's GLM-4.6 and CogView-4 through a credential-free bridge.

GITHUB.COM/20KIWY/ARTCRAFT-EGUI · APPS/ARTCRAFT + APPS/ARTCRAFT-PROXY